LUCAS HAN

+44 7752 840022| lucashan1225@gmail.com
haojunhan.com | github.com/LucasH1225
Available July–September 2027

EDUCATION

University of Oxford **Oct. 2025 – Expected Jun. 2028**
St Catherine's College Oxford, UK

- BA Computer Science and Philosophy
  - Ranked in the top 10 in Computer Science and Philosophy Prelims.

Shenzhen College of International Education

- Outstanding Cambridge Learner Award: Attained a perfect score in Mathematics.

Aug.2021 – Jun.2025
Shenzhen, China

TECHNICAL SKILLS

- Languages: Python, JavaScript, Swift.
- Frameworks and tools: PyTorch, NumPy, React, Three.js, SwiftUI, Electron, Webots, pytest and XCTest.
- Development practices: AI-assisted coding and automated testing.

INTERNSHIP

Shenzhen Huyang Intelligent Innovation Co., Ltd. **Jul. 2026 – Aug. 2026**
Robotics Algorithm Developer

- Developed a partially observable cleaning-robot simulator in Python, with procedural maps, contact-based obstacle discovery and reproducible evaluation.
- Implemented a PyTorch DQN planner with global/local CNN features, action masking and deterministic exploration to navigate unknown environments.
- Achieved 100% task completion across 900 benchmark maps and 150 additional unseen maps in discrete simulation.
- Integrated the planner with Webots and validated motion control and obstacle-navigation behaviour.

COMPETITIONS

- USACO Platinum division.
- Led a four-member team to a HiMCM Meritorious Award.
- BMO Merit Award; qualified for USAMO.

PROJECT EXPERIENCE

Wankai | 3D Physics Combat Game **Oxford Tripo Club**

- Built a playable 3D combat game from scratch using JavaScript, React, Three.js and Electron, with local multiplayer, split-screen cameras and character/weapon selection.
- Implemented a custom 180 Hz fixed-step simulation, contact-driven movement and collision impulses; added automated physics and gameplay tests.

Personal AI Assistant | Swift and SwiftUI

- Built a native macOS assistant and iOS prototype that turn natural-language messages into linked records in a user-controlled Obsidian vault;
- Use of built-in large language model to analysis the data and displayed them by designing UI
- Added source checks, duplicate prevention, edit-aware undo and conflict resolution.

Hackathon | Team Lead and Developer **Jul. 2023 – Present**

- In Xacademy 2023, developed a 2D plotted game on the topic of "psychopaths" and led the team to the Best Creative Award.
- In Oxford 2026, built Photo Fumble, a real-time multiplayer browser game using TypeScript, React, Node.js, Socket.IO and Supabase, featuring five game modes based on players' photos, audio recordings and chat memories.

LEADERSHIP & ACTIVITIES

SCIE DEV Club **Sep. 2023 – Jun. 2025**
Club Leader Shenzhen, China

- Taught game and web development.
- Helped maintain ScieToday, the school app for accessing organised website information.